Roll No. ....

TEC–601

B. TECH. (ECE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

WIRELESS COMMUNICATION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Verify that the cluster size is $N = i^2 + j^2 + ij$. (CO1)

(b) Find SIR at the boundary of a cell in case of 7, 9, and 12 cells in a cluster. If the minimum required SIR for acceptable voice quality is 18 dB then which one cannot be preferred. Assume path loss exponent to be 4. (CO1)

(c) How cell splitting is performed ? Why can it increase the system capacity ? (CO1)

2. (a) Consider a transmitter operating at 1800 MHz transmits 4-watt power. Assume path loss exponent to be 4, shadow effect of 10.5 dB and the power at reference point (d_0 = 100 m) is – 32 dB. Find the received power at a distance of 3 km from transmitter and the allowable path loss. (CO2)

P. T. O.

(b) Derive an expression to represent the received signal using free space path loss model. (CO2)

(c) (i) Find the Fraunhofer distance for a transmitting antenna having the maximum dimension of 1m operating at 900 MHz frequency.

(ii) Define Fraunhofer region. (CO2)

3. (a) What is the difference between rms delay spread and maximum excess delay for the multipath power delay profile ? Explain. (CO3)

(b) Explain the terms : Doppler shift and coherence time. Use suitable mathematical derivations to explain. (CO3)

(c) Explain different types of small scale fading using suitable diagrams. (CO3)

4. (a) How does selection combining function ? Derive an expression to represent the average SNR improvement realized in Selection Combining. (CO4)

(b) Derive an expression to represent the average SNR improvement realized in Maximal Ratio Combining. (CO4)

(c) With help of suitable mathematical derivations, explain Rayleigh Fading Model. (CO4)

5. (a) State and explain the properties of maximum length sequence. (CO5 & CO6)

(b) Discuss the processes involved in direct sequence spread spectrum (DSSS) and frequency hop spread spectrum (FHSS). (CO5 & CO6)

(c) Explain CDMA and discuss its salient features. (CO5 & CO6)

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TEC–602

B. TECH. (ECE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

MICROWAVE ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is meant by dominant mode in a waveguide ? Explain with respect to TM wave propagation in a rectangular waveguide. (CO1)

   (b) A square waveguide (a by a) can propagate only TE_{10} and not TE_{11} or higher modes. In order to achieve this, what must be the size a ? (CO1)

   (c) An air-filled rectangular waveguide has a cross section of 6 cm x 4 cm.

   (i) Calculate the cutoff frequency of the dominant mode.

   (ii) Determine how many modes are passed at three times cutoff frequency of dominant mode. (CO1)

2. (a) Draw the schematic of a TWT device and discuss its working principle. (CO3)

(b) Explain the negative resistance region of operation in a tunnel diode. (CO3)

(c) Discuss the different modes of operation in a Gunn diode and mechanism of formation of dipole domain. (CO3)

3. (a) Draw and explain Magic TEE. Write down its S-matrix. (CO2)

(b) What is the need of a directional coupler ? Explain the parameters associated with a 4-port directional coupler when port-1 excited. (CO2)

(c) 40mW microwave signal is fed to arm 3 of E plane tee by a match terminator. Arms 1 and 2 are terminated respectively in 40 Ω and 60L loads and the wave impedence of the guide is 50Ω. Find the power delivered to the loads. (CO2)

4. (a) Explain the mechanism of S-Parameter measurements for a 3-port microwave network. (CO4)

(b) Write the scattering matrix for ideal three port circulator. A three port circulator has insertion loss of 3 dB, Isolation 30 dB and VSWR is 1.2. Find its scattering matrix. Assume the phases of all coefficient are zero. (CO4)

(c) Explain spectrum analyzer with the help of a block diagram. (CO4)

5. (a) Discuss Impedance and Frequency Scaling technique used for filter transformation. (CO5, CO6)

(b) Explain filter implementation technique using Kuroda identities. (CO5, CO6)

(c) Write short notes on the following : (CO5, CO6)

(i) RFID

(ii) MMIC

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TEC–603

B. TECH. (ECE) (SIXTH SEMESTER)

END SEMESTER EXAMINATION, June, 2023

VLSI TECHNOLOGY AND DESIGN

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is Integrated Circuit? Explain advantages of ICs. (CO1)

(b) What do you understand by electronic grade silicon, and CZ crystal growth technique ? (CO1)

(c) Explain classifications of ICs based on structure. (CO1)

2. (a) Explain Ion stopping and implantation techniques. (CO2)

(b) Explain Physical vapor deposition technique for metallization applications in VLSI. (CO2)

(c) Explain in brief about Annealing and its applications in VLSI. (CO2)

P. T. O.

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TEC–604

B. TECH. (ECE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

DATA COMMUNICATION NETWORKS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Write short notes on the following : (CO1)

   (i) Circuit Switching

   (ii) Direct Sequence Spread Spectrum

   (b) Differentiate between Guided and Unguided transmission medium. Also state their types. (CO1)

   (c) Ten sources, with a bit rate of 200 kbps each are to be multiplexed using TDM with no synchronizing bits. Answer the following questions about the final stage of the multiplexing : (CO1)

   (i) What is the size of a frame in bits ?

   (ii) What is the frame rate ?

   (iii) What is the duration of a frame ?

   (iv) What is the data rate ?

P. T. O.

2. (a) What are hamming codes ? Discuss its limitations. A dataword of at least 7 bits is required. Calculate values of k and n that satisfy this requirement. (CO2)

   (b) What is burst error ? What is the maximum effect of 2-ms burst of noise on data transmitted at the following rates ? (CO2)

   (i) 1500 bps

   (ii) 100 kbps

   (iii) 100 Mbps

   (c) What is Point-to-point protocol ? Explain its working and also state its frame format in detail. (CO2)

3. (a) Write the functions of Media Access Control sub layer. Compare pure ALOHA and slotted ALOHA. (CO3)

   (b) Explain IEEE 802.5 token ring protocol with the help of diagram and frame format. (CO3)

   (c) A slotted ALOHA network transmits 200 bit frames on a shared channel of 100 Kbps. What is the vulnerable time ? Calculate the throughput if the system produces : (CO3)

   (i) 1000 frames/sec

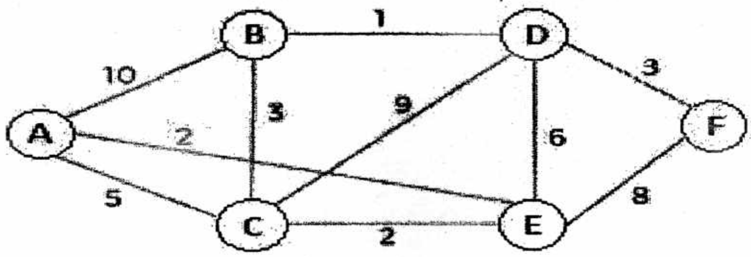
   (ii) 500 frames/sec

   (iii) 250 frames/sec

4. (a) Explain User datagram protocol (UDP). Discuss its frame format, working and mention its applications. (CO4)

   (b) Discuss in detail the steps involved in Distance vector routing algorithm with the help of diagram. (CO4)

(c) Consider the network shown in the picture below. Show the step-by-step process in computing the shortest path from source node A to all other nodes using Dijkstra's routing algorithm. (CO4)



5. (a) Write short notes on the following protocols : (CO5, CO6)

   (i) Domain Name System

   (ii) Simple Mail Transfer Protocol

   (b) Explain Multiprotocol Label Switching (MPLS). (CO5, CO6)

   (c) Explain cryptography and its types. Compare between Substitution and Transposition Ciphers. (CO5, CO6)

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TEC–659

B. TECH. (ECE) (SIXTH SEMESTER)

END SEMESTER EXAMINATION, June, 2023

ADVANCED EMBEDDED SYSTEM

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the different applications of Embedded Systems and General Computing Systems. (CO1)

(b) What are Quality Attributes of Embedded system ? (CO1)

(c) An embedded system-based visitor counter is needed. The system must be designed such that it can count only upto 100 visitors and after than it must blink the RED led. Explain the Embedded System design process for such system with the help of proper block diagram, flow chart for algorithm etc. (CO1)

2. (a) Explain the PIC microcontroller with the help of its features and architectures. (CO2)

   (b) Explain 8051 internal architecture and features of 8051 microcontrollers. (CO2)

   (c) Explain the timers of 8051. What is the role of Gate bit ? Discuss the details of registers used for Timer of 8051. (CO2)

3. (a) Explain Sensor interfacing with 8051. How ADC and Digital sensors are utilized for this purpose ? (CO3)

   (b) A LED 7-segment display is connected to P0. Write a program to count from 0 to 9 with specific interval of 1 sec. (CO6)

   (c) Explain Programming framework in Embedded Systems design. (CO3)

4. (a) What is difference between multiprocessing and multitasking ? (CO4)

   (b) What is the role of Real time operating systems (RTOS) in Embedded system design ? (CO4)

   (c) What is Task scheduling ? (CO4)

5. (a) What are AVR family microcontrollers? Explain ATMEGA16 microcontroller with the help of its architecture. (CO5)

   (b) Draw the Architecture of 8096 and explains its blocks in details. (CO5)

   (c) What is AMBA bus in ARM 32 microprocessor? Explain in detail with the help of diagram. (CO6)

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Roll No. ....

TOE–610

B. TECH. (ECE) (SIXTH SEMESTER)
END SEMESTER EXAMINATION, June, 2023

OOPS USING C++

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is call by value and call by reference in C++ ? How does the class accomplish data hiding in C++ ? (CO1, CO2, CO3)

   (b) Write a C++ program to create a class called following members : (CO1, CO2, CO3)

   Private Data Member :

   string str

   Public Function Members :

   void readValue() → to accept a sentence.

   void freqVowel() → Counts and print the frequency of the words that begin with a vowel.

   In main function create object of class VowelWord and perform the above task.

P. T. O.

(c) Differentiate between a constructor and a destructor. Write a program that demonstrates the use of default arguments in a constructor.

(CO1, CO2, CO3)

2. (a) List at least five operators that can be overloaded. Give the implementation of any *two* of them in detail. (CO1, CO3, CO4)

(b) Describe multipath inheritance C++ ? Provide the solution to the diamond problem with a C++ code. (CO1, CO3, CO4)

(c) Explain the process of opening a file in C++. Write a C++ program that exchanges the contents of two files. (CO1, CO3, CO4)

3. (a) Explain following with help of C++ program : (CO3, CO4, CO6)

(i) new and delete operator

(ii) this pointer

(b) What are function templates ? Write a C++ program to swap two variables of any type using function templates. (CO3, CO4, CO6)

(c) In multiple inheritances, what is the order of calling the constructors ? Explain with a sample C++ code. (CO3, CO4, CO6)

4. (a) How is run-time polymorphism achieved in C++ ? Explain pure virtual function with help of C++ program. (CO1, CO2, CO4)

(b) Write a C++ program to merge two integer arrays. Also display the merged array in reverse order. (CO1, CO2, CO4)

(c) Create an abstract class FactDemo with the following pure virtual function : (CO1, CO2, CO4)

virtual void factorial(int)=0;

Create a derived class Calculation which inherit abstract class FactDemo and override the function.

void factorial(int) method will calculate factorial of a passing number.

5. (a) What are the Exception Handling Keywords in C++ ? Write a C++ program that throws an exception of index out of bound.

(CO1, CO2, CO5)

(b) What are static data members ? Why are they needed ? When do we pass constant objects as parameters ? (CO1, CO2, CO5)

(c) Write a C++ program to read a file that contains small case characters. Then write these characters into another file with all lower case characters converted into upper case. (CO1, CO2, CO5)